Ethan Chang

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OBJECTIVE

Computer Science B.S. student (May 2027) seeking full-time Software Engineering, Data Engineering, Data Science, or AI roles.

EDUCATION

Rochester Institute of Technology **Expected May 2027**
Bachelor of Science, Computer Science; GPA: 3.6/4.0 *Rochester, NY*
Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C, C#, R, HTML/CSS
Frameworks & Libraries: React, Node.js, Express, Angular, Spring Boot, Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, JUnit
AI & LLMs: Claude API, DeepSeek-R1, Prompt Engineering, Text Embeddings
Tools & Platforms: Git, GitHub Actions, Docker, PostgreSQL, Linux, Maven, Jupyter Notebook, DataGrip, Vercel, Render

EXPERIENCE

Sandia National Laboratories **May 2026 – Present**
R&D Intern *Albuquerque, NM (Onsite)*

- Built a training tool that simulates data from hypothetical radioactive material releases, partnering with laboratory scientists to translate nuclear-science data into software features.
- Earned a **Thunderbird Award** (Purpose-Driven category) for contributions during the internship.

Artificial Intelligence and Biomedical Informatics and Data Science (AIBIDS) Program **May 2025 – Jul 2025**
Data Science Intern *Pittsburgh, PA (Hybrid)*

- Surfaced risk patterns in **5,000+** de-identified patient records to inform a predictive model for unplanned surgery.
- Extracted **464** pre-surgery clinical notes from a UPMC ACLR database (**2013–2021**) using SQL CTEs, joins, and embeddings.
- Classified activity levels with an LLM (DeepSeek-R1 14B), reaching **0.89** F1 on elite athletes via **6** hand-annotated example batches.
- Built logistic regression models in scikit-learn to predict unplanned surgery from social determinants of health (SDoH).
- Presented a research poster on activity level as an SDoH predictor to faculty, peers, and stakeholders.

NTID Learning Center **Sep 2025 – May 2026**
Software Development Tutor *Rochester, NY (Onsite)*

- Built students' coding confidence by teaching debugging and computational thinking in 1:1 and group sessions.
- Made Python, Java, and Excel accessible to Deaf and hard-of-hearing students by teaching in ASL and spoken English.

PROJECTS

PlayIQ: Sports Analytics Dashboard | *React, Node.js, Python, XGBoost, Claude API* **Mar 2026 – Present**

- Built a React dashboard for **6** leagues (MLB, NBA, WNBA, NFL, NHL, NCAAB) with per-game matchup, form, and roster tabs.
- Engineered Log5, Poisson, and Binomial projection models; automated weekly XGBoost money-line retraining via GitHub Actions.
- Served live ESPN and MLB data via a cached, dependency-free Node.js backend with Claude API game analysis.

Secure Distributed Messenger | *C#, .NET, TCP/UDP Sockets, Cryptography* **Jan 2026 – Apr 2026**

- Built a peer-to-peer encrypted messenger with a team of **4**, using AES-256 keys exchanged via RSA-2048 and signed messages.
- Owned peer discovery and fault tolerance: UDP broadcast discovery, **15**-second heartbeat timeouts, and backoff reconnection.

Chroma: U-Fund Crowdfunding Platform | *Angular, TypeScript, Java, Spring Boot* **Aug 2025 – Dec 2025**

- Built a full-stack charity crowdfunding platform with a team of **4**: **15** Angular components over a Spring Boot REST API.
- Designed a layered controller/model/DAO architecture with swappable persistence, covered by **15** JUnit test classes.

Movie Database Management System | *Python, PostgreSQL, SQL, SSH* **Jan 2025 – May 2025**

- Built an ETL pipeline with a team of **5** that cleans Kaggle datasets and loads them into a normalized PostgreSQL schema.
- Developed a modular Python application with SSH tunneling for secure remote database access and management.

LEADERSHIP & INVOLVEMENT

RIT/NTID Cross-Registered Student Advisory Board (XSAB) **Sep 2025 – May 2026**
Golisano College of Computing and Information Sciences (GCCIS) Representative *Rochester, NY*

- Represented cross-registered students within the RIT/NTID community, sharing XSAB updates on events, resources, and services.
- Selected by the NTID Co-op and Career Center as NTID Career Fair Ambassador in 2025 and 2026.